Entergy Operations, Inc.
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Robert Franssen
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10 CFR 50.73

GNRO-2021/00029

August 16, 2021

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2021-002-00, Core Monitoring System Software Error Resulted in Conditions Prohibited by Technical Specification

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report 2021-002-00, Core Monitoring System Software Error Resulted in Conditions Prohibited by Technical Specification. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), as an operation or condition which was prohibited by the plant's Technical Specifications.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact Jeff Hardy, Regulatory Assurance Manager at 269-764-2011.

Sincerely,

Robert Franssen
RF/fas

Attachments: Licensee Event Report 2021-002-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Attachment
Licensee Event Report 2021-002-00

 U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name	2. Docket Number	3. Page
Grand Gulf Nuclear Station, Unit 1	05000 - 416	1 of 3

4. Title
Core Monitoring System Software Modeling Error Resulted in Conditions Prohibited by Technical Specifications

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
									N/A	05000 N/A
06	17	2021	2021	-002 -	00	08	16	2021	Facility Name N/A	Docket Number 05000 N/A

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(4)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(1)(i)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(iii)	**10 CFR Part 50**	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ **Other** (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include Area Code)
Jeff Hardy, Manager Regulatory Assurance	(269)-764-2011

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☒ No	☐ **Yes** (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On June 17, 2021, with Grand Gulf Nuclear Station (GGNS) operating at 100 percent power, Entergy received a letter from GE Hitachi issuing SC 21-04, Revision1, “Fuel Support Side Entry Orifice Meta-Stable Flow for 2 Beam Locations in the BWR/6 Reactors,” identifying that the Global Nuclear Fuels (GNF) model for fuel affected by the Side Entry Orifices did not accurately account for the loss coefficients at those core locations causing a local overprediction in Minimum Critical Power Ratio (MCPR) margin. This resulted in the determination by Entergy that a condition prohibited by Technical Specifications (TSs) had existed at GGNS and this report being made in accordance with 10 CFR 50.73(a)(2)(i)(B).

The cause of the event was GNF had not accurately accounted for the loss coefficients at applicable core locations causing a local overprediction in MCPR margin resulting in exceeding the MCPR TS limit.

Corrective actions include implementing a penalty to Operating Limit MCPR (OLMCPR) in the core monitoring system via an update to the databank to include all penalties associated with GEH letter SC 21-04, Revision 1, and an update to the current Core Operating Limits Report (COLR).

There were no consequences to the general safety of the public, nuclear safety, industrial safety, nor radiological safety.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Grand Gulf Nuclear Station, Unit 1	05000 - 416	2021	- 002	- 00

NARRATIVE

Plant Conditions:
Grand Gulf Nuclear Station (GGNS) Unit 1 was operating at 100 percent power in MODE 1. There were no Structures, Systems, or Components that were inoperable that contributed to this event.

Event Description:
On June 17, 2021, with GGNS operating at 100 percent power, Entergy received a letter from GE Hitachi (GEH) concerning a Part 21 Report. This Part 21 Report was transmitted to Entergy and the NRC documenting an update on the issue related to an underprediction of the Side Entry Orifice (SEO) loss coefficients in core analyses, resulting in a local overprediction in MCPR margin.

Specifically, Global Nuclear Fuels (GNF) identified the SEO loss coefficients for fuel bundle locations adjacent to 2-beam (corner) locations may potentially be higher than the current design basis value that is applied. The MCPR impact on potentially affected bundles that are near limits can potentially reduce this margin by greater than 0.01 in Critical Power Ratio. GEH has completed the evaluation of the condition to determine reportability under 10 CFR Part 21 and is a reportable condition under 10 CFR Part 21.21(a)(2) and a transfer of information under 10 CFR 21.21(b). The basis for reportability is that the change in MCPR associated with this issue could contribute to the exceeding of a Safety Limit, as defined in the GGNS TSs.

The history of variable Maximum Fraction of Limiting Core Power Ratio (MFLCPR) for the last 3 years was reviewed. The objective was to identify any period where the MFLCPR (corrected for SC 21-04) exceeded 1.0 for more than 6 hours. During Cycle 22 (August 2018 through January 2020) and Cycle 23 (2020 – Present) there were multiple periods where MFLCPR was greater than 1.0 for longer than 6 hours. During these instances, the requirements of TS 3.2.2 were not met; however, during this time period there were no limiting transients that would be a challenge to the Safety Limit and there were no Safety Limit violations.

GGNS previously performed an Op Eval for SC 21-04 Revision 0 in CR-GGN-2021-3024 CA-01, which resulted in the implementation of a compensatory measure for tracking CR-GGN-2021-04909. To ensure compliance until the new COLR is approved, the site issued a standing order for reduced administrative MFLCPR as the compensatory measure. The site's review of SC 21-04 Revision 1 was performed and based on that review; the standing order was updated to add additional conservatism.

The BWR/6 plant design is that it has supporting cross beams that form a grid structure underneath the core plate. The orientation of SEO's relative to the beams produces the different losses due to difference in upstream flow areas. These variations influence the SEO flow patterns and the potential for a meta-stable pressure loss. Because the frequency at which meta-stable losses may occur has not been determined, GGNS was evaluated using a bounding loss coefficient (1.9 times the current loss value) at the 2-beam location. Results show the potential MCPR impact at limiting locations can be greater than 0.01 and will vary by plant and power/flow condition. The MCPR impact is greater than the 0.01 criterion that GEH has historically applied for reporting that a TS Safety Limit (as found in TS 2.1) could have been exceeded as defined under 10 CFR Part 21. If the MCPR impact is greater than 0.01 due to possible meta-stable losses at the 2-beam locations, and if not addressed, the condition could occur at a limiting bundle location and reduce transient margin. This could have resulted in exceeding the Safety Limit Minimum Critical Power Ratio (SLMCPR) if a limiting transient were to have occurred.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
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Safety Assessment:
There were no actual consequences for this event. The GGNS Operating Limit MCPR (OLMCPR) is set such that if the plant is operating at this limit, the most limiting operational transient for the cycle will not result in violation of the SLMCPR. The site reviewed the unit operating history for limiting transients and other analyzed transients that are close to being limiting in the Supplemental Reload Licensing Reports. This review concluded that, for the period reviewed, the limiting transients were turbine trip without bypass, feedwater controller failure, generator load reject without bypass, rod withdrawal error, and loss of feedwater heating.

Because these limiting transients were not identified in the period reviewed, the margin between the SLMCPR and OLMCPR was adequate for continued SLMCPR protection. The SLMCPR was not exceeded in previous analyzed cycles and the margin between SLMCPR and OLMCPR was sufficient to accommodate the penalty evaluated in SC 21-04, Revision 1. As a result, there was no impact to the health and safety of the public or plant personnel from this condition. In addition, this event does not meet the criteria for a Safety System Functional Failure.

Event Cause(s):
Based on the Entergy internal review, it was determined that the cause of this event was: GNF did not accurately account for the loss coefficients at those core locations causing a local overprediction in MCPR margin resulting in exceeding the MCPR TS limit.

These occurrences of exceeding the MCPR TS limit were beyond the control of Entergy to predict or prevent because it is a legacy design calculation error. Entergy does not have the capability to check these calculations, nor does Entergy have the resources to perform an independent review of all GNF calculations. The methodology used in the reload process has been accepted by Entergy and we do not check underlying calculations that have existed unchanged for multiple cycles. Entergy will credit actions taken by GNF as part of the Part 21 notification, including actions they are taking for extent of condition in their calculations. This would include the results of any technical audit they are performing. Entergy Supplier Quality Assurance was notified of this Part 21 for follow-up on the GNF corrective action.

Corrective Actions:
To ensure compliance with the MCPR TS, GGNS issued the standard order as described above. Complete.

GGNS has implemented a penalty to OLMCPR in the core monitoring system through an update to the databank to include all penalties associated with GEH letter SC 21-04, Revision 1. Complete.

GGNS will update the current COLR and associated documents to account for all penalties associated with GEH letter SC 21-04, Revision 1. Completion tracked by Corrective Action Program.

Entergy's Supplier QA has added CR-HQN-2021-01048 and SC 21-04 R1 to the Qualified Supplier List comments sections for GEH/GNF Fuels for follow-up in the next audit. Complete.

Previous Similar Events:
None.